Quantum Machine Learning Methods for Semiconductor Wafer Defect Classification

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Introduction

Defects introduced during semiconductor wafer fabrication can reduce yield, making rapid and accurate defect detection essential for quality control. Automated defect classification systems are increasingly using machine learning techniques to analyze wafer maps and identify patterns of faulty dies [1]. At the same time, quantum machine learning (QML) algorithms have been proposed to exploit high-dimensional quantum feature spaces and parameterized quantum circuits for data analysis [2][3]. While QML methods have shown promise on small synthetic datasets, their performance on realistic industrial datasets remains largely unexplored. This project investigates whether quantum machine learning classifiers can be applied to the problem of semiconductor wafer defect classification.

Dataset

Three different QML models will be trained on the MixedWM38 WaferMap dataset [4][5], a publicly available dataset of semiconductor wafer maps used in defect detection research. This dataset contains 38,000 wafer images represented as 52×52 grids, with each pixel corresponding to a die on the wafer. Pixel values indicate whether the die is absent, functional, or defective, as seen in Figure 1. Many wafers exhibit spatial patterns of failure which can take several characteristic forms, including: (1) center defects, (2) donut, (3) edge-localized, (4) edge-ring, (5) localized, (6) nearly full wafer failure, (7) scratches, and (8) random. Each wafer is labeled using an 8-dimensional Boolean vector indicating which defect patterns are present. The goal of this project is to develop QML models capable of correctly identifying these defect patterns.

Proposed Methodology

Because each wafer image contains $52 \times 52 = 2704$ pixels, directly encoding the raw data into a quantum circuit is not feasible with current quantum hardware. To address this limitation, classical dimensionality reduction techniques like Principal Component Analysis (PCA) will be used to create a low-dimensional feature space containing approximately 10–15 principal components. This allows the dataset to be encoded into quantum circuits with a manageable number of qubits while preserving most of the relevant variance in the data.

Two quantum machine learning models will be implemented and evaluated. The first model is a **quantum kernel support vector machine** (QSVM). In this approach, a quantum circuit defines a feature map that embeds classical input data into a high-dimensional Hilbert space. The similarity between two data points is computed using the quantum kernel $K_{ij} = |\langle \phi(x_i) | \phi(x_i) \rangle|^2$ which measures the overlap between quantum states generated by the feature map. The resulting kernel matrix is then used by a classical SVM to determine the optimal decision boundary. The quantum circuit defining the feature map is fixed, and learning occurs during the classical SVM optimization.

The second model is a **variational quantum classifier** (VQC). In this approach, a parameterized quantum circuit acts as a trainable model analogous to a neural network [6]. Classical data is encoded into quantum states, after which a sequence of parameterized gates is applied. The circuit parameters are optimized using gradient-based methods to minimize a classification loss function. Because the wafer dataset is multi-label, the classification task will be trained on separate binary classifiers for each defect type ("one-vs-rest" ML).

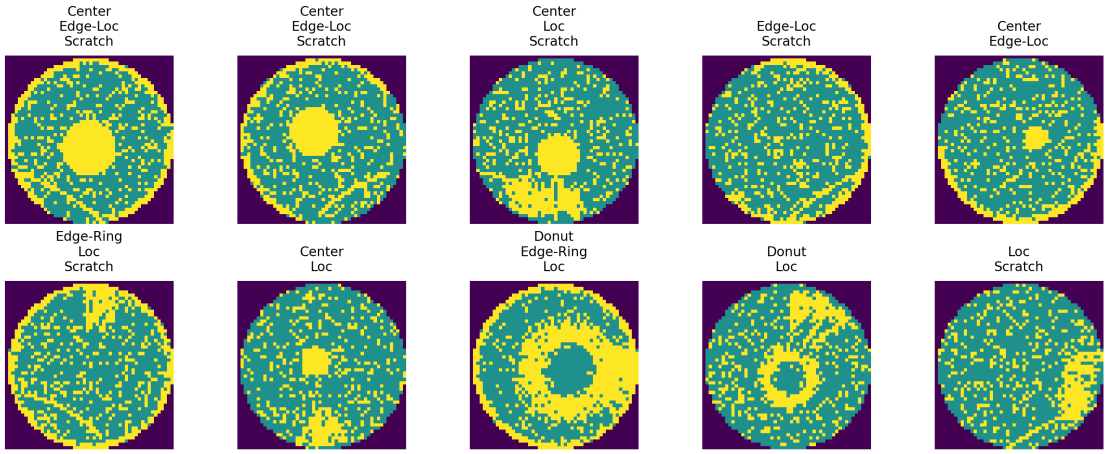

Figure 1: Examples of MixedWM348 semiconductor wafer images. Each image is 52 x 52 pixels, and each pixel represents a die that is functional (green), defective (yellow), or absent (purple). The macroscopic defect patterns are labeled above each wafer image.

Both models will be implemented using Python-based QML frameworks, specifically Qiskit Machine Learning [7] and PennyLane [8]. Quantum circuits will be executed on classical quantum simulators. Model performance will be evaluated using standard classification metrics such as accuracy, F1 score, and ROC-AUC, and results will be compared against classical SVM and convolutional neural net (CNN) baselines. By applying QML methods to a realistic semiconductor manufacturing dataset, this project aims to explore whether quantum feature maps or variational circuits can realistically be used for identifying spatial defect patterns in wafer maps.

AI Disclaimer: ChatGPT and Gemini were used to organize research notes into an early draft of this report, and Claude to develop Python code for reading MixedWM348 data.

References

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